

附件四：加分佐證文件

「智慧創新大賞」加分佐證文件

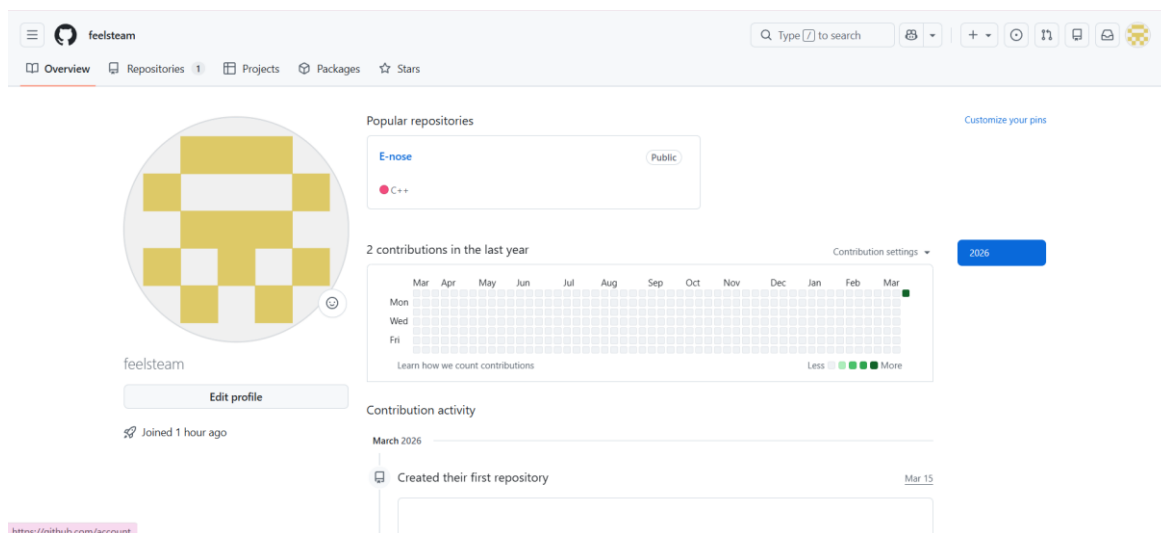
1. GitHub：

帳號：feelsteam

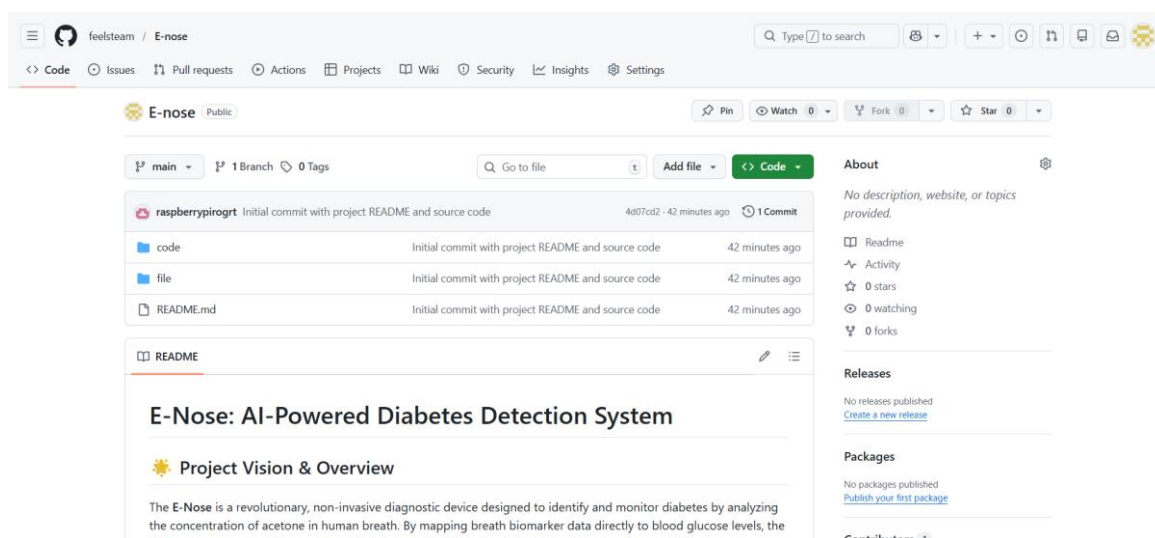
倉庫：E-nose

連結：<https://github.com/feelsteam/E-nose>

專案已全數上傳 github，並有撰寫專案介紹的 Readme.md 文件，其中對了 AI 應用有做特別撰寫，相關截圖如下：



圖一:帳號截圖



圖二:專案倉庫截圖

Core Highlight: Artificial Intelligence & Machine Learning

At the heart of the E-Nose system lies its sophisticated AI engine. Moving beyond simple threshold-based detection, we heavily leverage machine learning across multiple layers of the application to overcome the inherent challenges of gas sensing and physiological variability.

1. Dynamic Sensor Calibration Pipeline

Gas sensors exhibit non-linear responses and are highly sensitive to environmental factors (temperature, humidity, cross-sensitivity with other gases).

- **Neural Network Calibration:** We employ Deep Neural Networks (DNNs) to process multi-dimensional sensor data. The AI dynamically compensates for drift and environmental noise, mapping complex semi-logarithmic sensor curves into hyper-accurate parts-per-million (ppm) readings of acetone.

2. Predictive Medical Analytics

- **Glucose Level Correlation:** The core AI models are trained on extensive clinical datasets matching breath acetone levels to precise venous blood glucose concentrations.
- **Early Warning System (EWS):** Utilizing time-series forecasting models (like LSTMs or Transformers), the system learns a patient's historical trends to predict impending hyper/hypoglycemic events *before* they manifest physically, alerting the user proactively.

3. Personalized Physiological Baselines

Every patient's metabolism is unique. What constitutes a high acetone level for one individual may be completely normal for another.

- **Federated Learning & Adaptation:** The E-Nose AI continually fine-tunes itself to the individual user. It establishes a personalized baseline over the first few weeks of use, dramatically improving false-positive/false-negative rates over time without compromising patient data privacy.

圖三:有關 AI 的 Readme 敘述

2. Stack Overflow :

帳號：team feels

問題：有關感測器在不同溫濕度下的啟動誤差的解決辦法

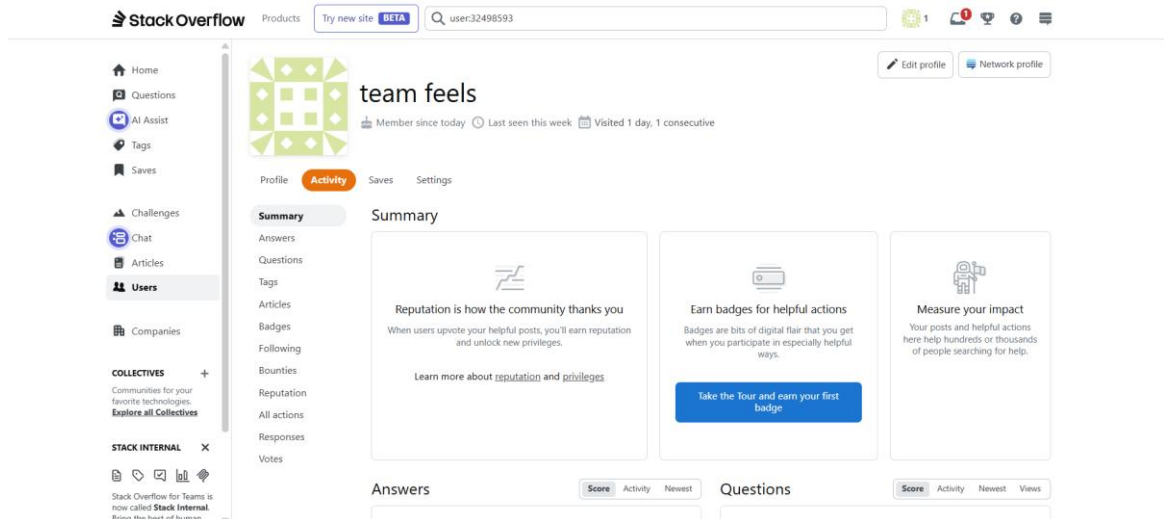


圖 4:帳號截圖

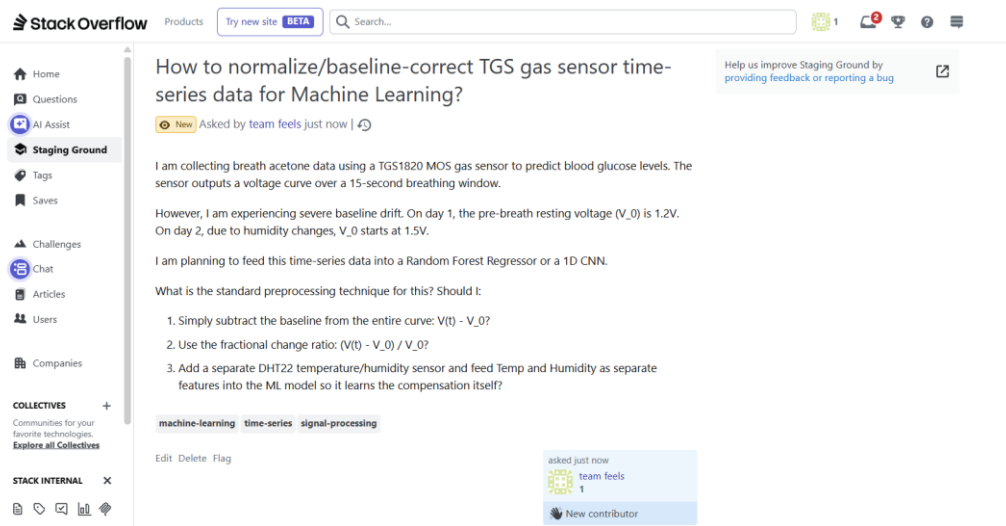


圖 5:提問截圖

3. Hugging Face :

帳號：feelsteam

倉庫：E-nose

連結：<https://huggingface.co/feelsteam/E-nose>

上傳雲端訓練用模型，相關截圖如下：

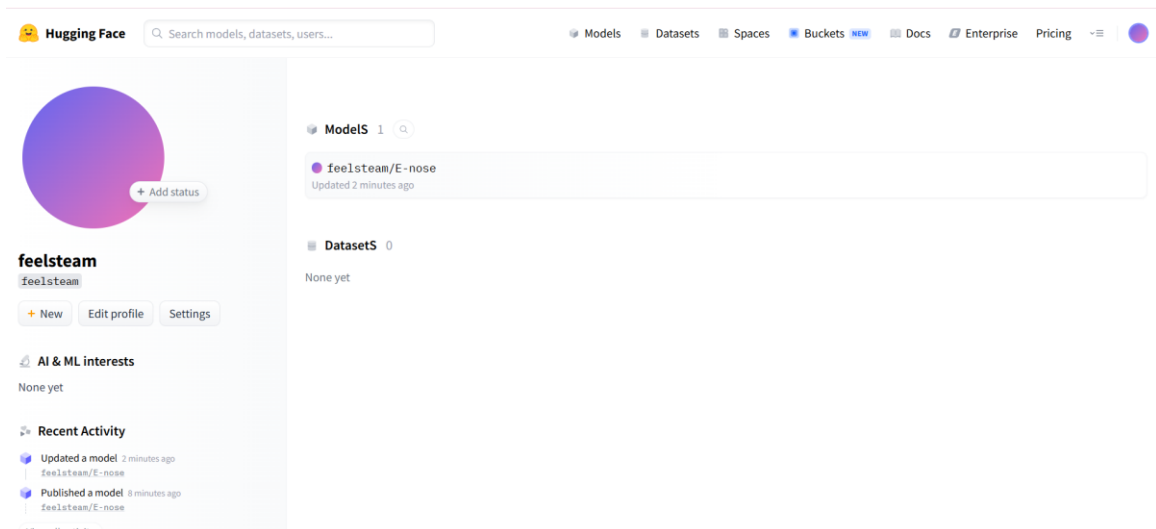


圖 6:帳號截圖

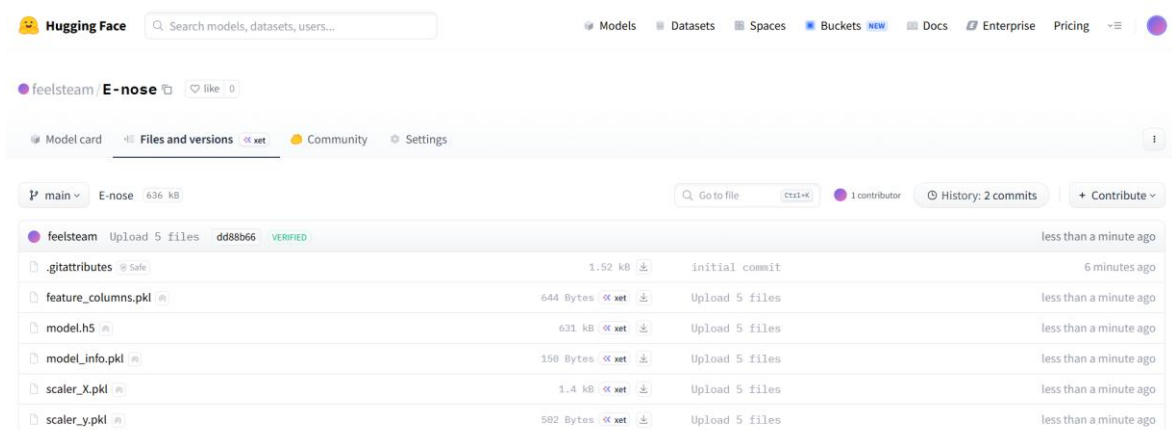


圖 7:倉庫(模型)